

ÁLVARO ROMERO CASTILLO

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📍 Dublin, Ireland / Barberà del Vallès, Spain

EDUCATION

Agora Sant Cugat International School

Primary school, Secondary education and International Baccalaureate diploma.

2007–2022

Grade: 39/45

Subjects:

- Physics HL (7), Mathematics HL (7), Spanish A HL, English B HL, Chemistry SL, History SL.

Trinity College Dublin

B.A. Honours Bachelor Degree in Theoretical Physics. Predicted: First Class Honours.

2022–present

Dublin, Ireland

- **Final year subjects:** Quantum Field Theory; Problem Solving; Applied Differential Geometry; Lie Groups, Lie Algebras and Physics; Interacting Quantum Systems; Quantum Optics and Information; The Standard Model of Elementary Particle Physics.

AWARDS

Bojos per la física

A prestigious physics programme in which they select 25 high school students from Catalonia to visit local investigation centers such as the ICFO, IFAE, ALBA synchrotron; and gain early exposure to the current state of research in fields such as high energy physics, nanotechnology and astrophysics.

Fundació Catalunya La Pedrera,
Barcelona

2022

Arthur Lyster prize

Awarded to the best 8 results obtained in first grade in the Mathematics and Theoretical Physics cohort. In particular awarded the price for third place in the School of Mathematics, first in Theoretical Physics.

School of Maths,
Trinity College Dublin

2023

DAAD RISE Scholarship

Funding given by the German Federal Foreign Office via the German Academic Exchange Service (DAAD) to complete research at the University of Stuttgart during the summer period. In 2025, 2,512 North American, British, and Irish students applied for 250 scholarships.

Deutsche Akademischer
Austauschdienst

2025

RESEARCH EXPERIENCE

Stuttgart Center for Simulation Science (SimTech)

DAAD RISE Scholarship Intern

May-Aug 2025
Stuttgart, Germany

Paper in progress: Reservoir computing with non-physical delayed-input vs. physical active-matter response with Mario Gaimann and Miriam Klopotek.

- Worked under the supervision of PhD student Mario Gaimann in ResoBee, following the work of his [first paper](#) on reservoir computers.
- Developed many body system simulations in a high performant workstation with use of HOOMD in Python to test the robustness of the simulation and the readout layer, as well as tests to verify its behaviour using pytest.

Fourth Year Capstone Project

Trinity College Dublin, School of Mathematics

Sept 2025–Present
Dublin, Ireland

Title: Characterising the XY Model on the Lattice: Critical Exponents and Finite-Size Effects.

- Working on a research project under the supervision of Prof. Patrick Fritzsche.
- Computationally modelling the XY model on a 2D and 3D lattice using C++ to extract the physical exponents relevant to phase transitions and studying the effect of vortices in topological phase transitions; as well as lattice expansion and finite-size effects.

WORK EXPERIENCE

Alpha Options

Quantitative analyst

May 2024-May 2025
Barcelona, Spain

- Developed random forest machine learning models targeting high frequency scenarios in the stock market, specialising in options of index funds, to obtain positive return on investment.
- Managed large datasets with MongoDB and Docker to obtain the time series data with which to train and test the machine learning models.

SKILLS

Languages

Spanish: Native | **Catalan:** Native | **English:** Fluent | **German:** Basic

Programming

Languages: Python, C++, Julia, HTML, CSS.
Libraries: Numpy, Matplotlib, Pandas, Sympy, Scipy, Scikit-learn, Pixi, numba, pytest, HOOMD, Optuna.
Frameworks: Virtual Studio Code, MongoDB, Git, Docker, Vim, Linux, Windows, Nix, ssh.

EXTRACURRICULAR

Kumon A mathematics programme encompassing from counting to infinitesimal calculus that I completed with 13 years of age.	2008–2017 Sabadell, Spain
Model of the United Nations Conferences where High School students emulate the assemblies of the United Nations and debate on current world issues and try to reach solutions. I attended the Iberian Model of the United Nations in Lisbon 2019 and 2020, The Hague International Model of the United Nations on 2020 and 2021 and the National High School Model of the United Nations in New York on 2022.	2019–2022
Theoretical Physics Student Association Public relations officer during the beginning of 2024 and Admin officer during the 2024/25 term. Participated on the 2024 TPSA Hackathon that proposed to solve the population dynamics problem in Ireland, as well as on the Qiskit Fallfest on 2023 organised by IBM.	2024–2025 Dublin, Ireland
Dublin University Swimming Club Ordinary Committee Member during the 2023/24 year and Secretary of the club during 2024/25, as well as an active member.	2023–present Dublin, Ireland